Lecture 31

Tuesday, 13 October 2026 8:24 AM

Recap: Scalar field $f: \mathbb{R}^n \longrightarrow \mathbb{R}$

Vector field $f: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ $(m>1)$

Def: $\lim_{\bar{x} \to \bar{a}} f(\bar{x}) = L$

if $|f(\bar{x}) - L| \longrightarrow 0$ as $\|\bar{x} - \bar{a}\| \longrightarrow$

$\bar{x} = (x_1, x_2, \ldots, x_n)$, $\bar{a} = (a_1, a_2, \ldots a_n)$

Here $\|\bar{x} - \bar{a}\| = \sqrt{(x_1-a_1)^2 + (x_2-a_2)^2 + \cdots + (x_n-a_n)^2}$

Eg 1: $\lim_{(x,y) \to (0,0)} \frac{xy}{\sqrt{x^2+y^2}} = 0$? (Guess!)

$\left| \frac{xy}{\sqrt{x^2+y^2}} - 0 \right| \longrightarrow 0$ as $\|(x,y) - (0,0)\| \longrightarrow 0$?

$f(x,y)$ $\quad L$

ie $\left| \frac{xy}{\sqrt{x^2+y^2}} - 0 \right| \longrightarrow 0$ as $\sqrt{x^2+y^2} \longrightarrow 0$?

Consider $\left| \frac{xy}{\sqrt{x^2+y^2}} \right| \leq \frac{\left(\frac{x^2+y^2}{2}\right)}{\sqrt{x^2+y^2}}$

$\leq \frac{1}{2}\sqrt{x^2+y^2}$

As $\sqrt{x^2+y^2} \longrightarrow 0$,

$\left| \frac{xy}{\sqrt{x^2+y^2}} \right| \longrightarrow 0$

x^2, y^2

$AM = \frac{x^2+y^2}{2}$

$GM = \sqrt{x^2 y^2}$

$= |xy|$

$AM \geq GM$

$\frac{x^2+y^2}{2} \geq |xy|$

$|xy| \leq \frac{x^2+y^2}{2}$

Rem: $\lim_{\bar{x} \to \bar{a}} f(\bar{x}) = L$

Express $|f(\bar{x}) - L|$ in terms of $\|\bar{x} - \bar{a}\|$

Continuity of Scalar field

$f: \mathbb{R}^n \longrightarrow \mathbb{R}$.

$f: \mathbb{R}^n \to \mathbb{R}$ is Continuous at $\bar{a}$

if $\lim_{\bar{x} \to \bar{a}} f(\bar{x}) = f(\bar{a})$

$\bar{x} = (x_1, x_2, \ldots x_n)$

$\bar{a} = (a_1, a_2, \ldots a_n)$

Eg1:

$$f(x,y) = \begin{cases} \frac{xy}{\sqrt{x^2+y^2}} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

Is f Cont at $(x,y) = (0,0)$?

$f: \mathbb{R}^2 \to \mathbb{R}$

To prove that f is cont at $(0,0)$, we need to show $\lim f(x,y) = f(0,0)$

$(x,y) \to (0,0)$

// Proved

0

// Def

0

$\therefore$ f is cont at $(0,0)$.

Eg2:

$$f(x,y) = \begin{cases} e^{-y/x} & x \neq 0, \ y \in \mathbb{R} \\ 0 & x = 0, \ y \in \mathbb{R} \end{cases}$$

Is f Cont at $(0,0)$?

$\lim f(x,y) \overset{?}{=} f(0,0)$

$(x,y) \to (0,0)$

// Def

0

?

Doesn't exist

$e^{-y/x}$

Path 1: $y = x$, $x \to 0$ $\quad e^{-1}$

Path 2: $y = -x$, $x \to 0$ $\quad e^{+1}$

$\lim$

∴ The limit doesn't exist as it is depending on the path chosen.

∴ f is NOT cont at $(0,0)$.

Continuity of Composite function:

$$f(g(x,y))$$

$$f(g(x_1, x_2, \dots, x_n)) = f(g(\bar{x}))$$

Thm: If $g(\bar{x})$ is cont at $\bar{a}$ and f is cont at $g(\bar{a})$ then fog is cont at $\bar{a}$

Eg(1): Is $\log(x^2+y^2)$ cont at $(1,1)$?

$$f(g(\ )) = \log(x^2+y^2)$$

$f(z) = \log z$, $g(x,y) = x^2+y^2$

$$\Rightarrow fog(x,y) = \log(x^2+y^2)$$

Q1: Is $g(x,y)$ cont at $\bar{a} = (1,1)$?

$$\lim_{(x,y)\to(1,1)} g(x,y) = g(1,1)$$

i.e $\lim_{(x,y)\to(1,1)} x^2+y^2 \stackrel{?}{=} 2$

$\lim_{(x,y)\to(1,1)} x^2+y^2 = 2$

Q2: Is f cont at $g(\bar{a}) = g(1,1) = 2$?

Is $f(z) = \log z$ cont at $z = 2$?

Thm $\Rightarrow$ fog is cont at $\bar{a} = (1,1)$.

Eg2s! Discuss the continuity of $\log|(\cos(x^2+y^2))|$ at $(\sqrt{\pi}, 0)$?

$f : \mathbb{R}^n \to \mathbb{R}$ is cont at $\bar{a}$

if $\lim_{\bar{x} \to \bar{a}} f(\bar{x}) = f(\bar{a})$.

$|f(\bar{x}) - f(\bar{a})| \to 0$ or $\|\bar{x} - \bar{a}\| \to 0$

↳ norm

Express $|f(\bar{x}) - f(\bar{a})|$ in terms of $\|\bar{x} - \bar{a}\|$